

Stage I: Quantization-aware self-distillation

Prompts
 $x \sim \mathcal{D}$

Frozen BF16 teacher
sample y and logits

y , top- k logits

FP4 student
masked token KL

$$\mathcal{L}_{\text{KD}} = T^2 \sum_{t \in y} \text{KL}(p_T \| p_S)$$

Stage II: FP4 GRPO post-training

QAT-adapted
FP4 policy π_θ

Quantized rollouts
 $\{y_i\}_{i=1}^G$

Task rewards and
group advantages

Update policy
with GRPO

FP4 linear operator inside the policy

Weight path W
fixed $W \approx U\Sigma V^\top$
FP4 factors

Activation path X
 $X = \mathbf{1}\mu_X^\top + R_X$
quantize R_X in FP4

Gradient path G
 $G = \mathbf{1}\mu_G^\top + R_G$
quantize R_G in FP4

Causal activation mean during autoregressive decoding

Prompt prefix
compute μ_0, n_0

Causal cache
($\mu_{\text{cache}}, n_{\text{cache}}$)

Decode step t
update with μ_t, n_t